Geometry, Entropy, and Operator Learning with Tunable Kernels

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Contents

1 Introduction

1.1 Why tune local kernels?

1.2 Key contributions

- Closed-form bridge: CGF curvature → Fisher load → curvature & entropy.
- Single KL optimisation (little-rose) jointly tunes **two** dials (θ, σ).
- Finite-sample Goldenshluger–Lepski selector with risk guarantees.

1.3 Organisation of the paper

2 Rose kernels and dials

2.1 Product-form kernels and minimal assumptions

$$z := y - x, \qquad r := \|z\|.$$

(W1) (**Radial weight**) $w : [0, \infty) \to [0, \infty)$ is C^1, radial, $\int_{\mathbb{R}^d} w(\|z\|)\, dz = 1$, $\int \|z\|^2 w(\|z\|)\, dz < \infty$, and $w \in L^2(\mathbb{R}^d)$.

(W2) (**Centred residual**) $g : \mathbb{R}^d \to [0, \infty)$ is C^1, $\int g(z)\, dz = 1$, $\int z\, g(z)\, dz = 0$, $\int zz^\top g(z)\, dz < \infty$, and $\nabla g \in L^2(\mathbb{R}^d)$.

Definition 2.1 (Rose kernel)**.** For $(\theta, \sigma) \in (0, \infty)^2$ set

$$w_\theta(r) := \theta^{-d} w\big(r/\theta\big), \quad g_\sigma(z) := \sigma^{-d} g\big(z/\sigma\big), \quad k_{\theta,\sigma}(y \mid x) := \frac{w_\theta(\|x - y\|)\, g_\sigma(y - x)}{\displaystyle\int_X w_\theta(\|x - y'\|)\, g_\sigma(y' - x)\, dy'}$$

Lemma 2.1 (Normalisation and basic bounds)**.** *Under Assumptions (W1)–(W2) the expression in Definition 2.1 defines a measurable Markov transition density $k_{\theta,\sigma}(y \mid x)$:*

$$0 \le k_{\theta,\sigma}(y \mid x) \le \frac{w_\theta(\|x - y\|)\, g_\sigma(y - x)}{\inf_x Z_{\theta,\sigma}(x)}, \qquad \int_X k_{\theta,\sigma}(y \mid x)\, d\mu(y) = 1,$$

where $Z_{\theta,\sigma}(x)$ is the denominator in Definition 2.1 and satisfies $0 < \inf_x Z_{\theta,\sigma}(x) \le \sup_x Z_{\theta,\sigma}(x) < \infty$.

(K1) (**Markov**) $k_{\theta,\sigma}(y \mid x) \geq 0$ and $\int_X k_{\theta,\sigma}(y \mid x)\, dy = 1$.

(K2) (**Score in** L^2) $S_{\theta,\sigma}(y \mid x) := \nabla_y \log k_{\theta,\sigma}(y \mid x) \in L^2(dy\, dx)$.

(K3) (**Second-moment scaling**) $\displaystyle\int_X \|y - x\|^2 k_{\theta,\sigma}(y \mid x)\, dy = O(\theta^2)$ as $\theta \downarrow 0$.

(K4) (**C^1 in dials**) $\partial_\theta \log k_{\theta,\sigma}, \partial_\sigma \log k_{\theta,\sigma} \in L^2(dy\, dx)$.

2.2 Local–linear residual construction (canonical example)

2.3 Scaled kernels and notation for the rest of the paper

2.4 Neutral (reversible) Rose operator

- Define $k^\natural(y \mid x) = \frac{1}{2}[k(y \mid x) + k(x \mid y)]$.
- Self-adjoint $\mathcal{R}^\natural$; serves as equilibrium baseline.

3 Population theory

(K0)–(K2) as in Sec. 3.0 — Markov, score L^2, second-moment scaling.

3.1 CGF $\Rightarrow$ Fisher $\Rightarrow$ Landau

Definition 3.1 (One–step CGF)**.**

Definition 3.2 (Symmetric Fisher load)**.** Given a Rose kernel $k_{\theta,\sigma}(y \mid x)$ with score $S_{\theta,\sigma}(y \mid x) := \nabla_y \log k_{\theta,\sigma}(y \mid x)$ and anchor/target measures μ_0, μ_1, define for $f \in W^{1,2}(X, \mu)$

$$\mathcal{I}_{\theta,\sigma}[f] \;:=\; \iint_{X\times X} \big\langle \nabla f(y),\, S_{\theta,\sigma}(y \mid x) \big\rangle^2 \, d\mu_0(x)\, d\mu_1(y).$$

Similarly, for a density $\rho \ll \mu$,

$$\mathcal{I}_{\theta,\sigma}[\rho] \;:=\; \iint_{X\times X} \big\langle \nabla \log \rho(y),\, S_{\theta,\sigma}(y \mid x) \big\rangle^2 \, d\mu_0(x)\, d\mu_1(y).$$

Proposition 3.1 (CGF–Fisher correspondence)**.** *For every $(\theta, \sigma) \in (0, \infty)^2$ and every centred observable $f \in W^{1,2}(X, \mu)$,*

$$\Lambda_{\theta,\sigma}(\lambda; f) = \frac{\lambda^2}{2}\, \mathcal{I}_{\theta,\sigma}[f] + O(\lambda^3), \qquad \lambda \to 0,$$

where $\mathcal{I}_{\theta,\sigma}[f] = \iint \langle \nabla f(y), S_{\theta,\sigma}(y \mid x)\rangle^2 \, d\mu_0(x)\, d\mu_1(y)$.

Remark 3.1 (Legendre dual = Landau stiffness)*.* A two–line box reminding readers that the Legendre transform of Λ yields a quadratic Landau free energy.

3.2 Dirichlet energy as common source

Definition 3.3 (Symmetrised kernel and Dirichlet form)**.** Let $k^{\natural}_{\theta,\sigma}(y \mid x) := \frac{1}{2}\big(k_{\theta,\sigma}(y \mid x) + k_{\theta,\sigma}(x \mid y)\big)$ and let $\mu_\natural$ be any reference measure absolutely continuous w.r.t. both μ_0 and μ_1 (often $\mu_\natural = \frac{1}{2}(\mu_0 + \mu_1)$). Define the associated Dirichlet form

$$\mathcal{E}_{\theta,\sigma}(f) := \frac{1}{2} \iint_{X \times X} \big(f(y) - f(x)\big)^2 \, k^{\natural}_{\theta,\sigma}(y \mid x) \, d\mu_\natural(x) \, d\mu_\natural(y).$$

Lemma 3.1 (Fisher load from Dirichlet energy)**.** *Under Assumptions (K1)–(K3), there exists a dial-dependent coefficient $c_{\theta,\sigma} > 0$ such that*

$$\mathcal{I}_{\theta,\sigma}[f] \;=\; c_{\theta,\sigma} \, \mathcal{E}_{\theta,\sigma}(f) \;+\; o(1) \qquad (\theta \downarrow 0),$$

with $c_{\theta,\sigma} = 4/\theta^2 + o(\theta^{-2})$ in the isotropic small-locality regime.

Remark: Reversible baseline and operator split

3.3 Fisher–curvature–entropy bridge

Lemma 3.2 (Iterated carré–du–champ)**.** *Under Assumptions (K1)–(K3), for every $f \in W^{1,2}(X, \mu)$,*

$$\Gamma_2^{(\theta,\sigma)}(f) = \frac{\theta^2}{2} \, \mathcal{I}_{\theta,\sigma}[f] + o(\theta^2), \qquad \theta \downarrow 0.$$

Lemma 3.3 (Discrete de Bruijn Identity)**.** *Fix $\theta > 0$. For every density $\rho \ll \mu$ with $\mathcal{I}_{\theta,\sigma}[\rho] < \infty$,*

$$\frac{d}{d\sigma} H\big(R^{\sharp}_{\theta,\sigma}\rho\big) = -\frac{1}{2} \, \mathcal{I}_{\theta,\sigma}[\rho], \qquad \sigma > 0.$$

Theorem 3.1 (Fisher–curvature–entropy bridge)**.** *With Assumptions (K1)–(K3),*

$$\boxed{\mathcal{I}_{\theta,\sigma} = \frac{2}{\theta^2} \, \Gamma_2^{(\theta,\sigma)} = -2 \, \frac{d}{d\sigma} H\big(R^{\sharp}_{\theta,\sigma}\rho\big)} \;+\; o(1).$$

Corollary 3.1 (Discrete fluctuation–dissipation)**.** *Combine Lemma 3.2 and Lemma 3.3. In the small–locality regime $\theta \downarrow 0$, the response coefficient obtained by differentiating the one–step entropy $H(R^{\sharp}_{\theta,\sigma}\rho)$ w.r.t. the noise dial σ equals (up to the universal factor $2/\theta^2$) the curvature prefactor $\Gamma_2^{(\theta,\sigma)}(f)$, i.e.*

$$\frac{d}{d\sigma} H\big(R^{\sharp}_{\theta,\sigma}\rho\big) \;=\; -\frac{1}{\theta^2} \, \Gamma_2^{(\theta,\sigma)}(f) \;+\; o(1).$$

Thus fluctuations (Fisher load) determine dissipation (entropy production) in exact analogy with the Onsager reciprocal relations.

3.4 Discrete Onsager–Machlup functional

Corollary 3.2 (Discrete OM action)**.** *Let* $\mathbf{x} = (x_0, \dots, x_n)$ *be a trajectory sampled every* h *and let* $k_{\theta,\sigma}$ *satisfy* (K0)–(K2). *With local drift* m_θ *and quadratic forms* $\Gamma_t^2 := \Gamma_2^{(\theta,\sigma)}(f_t)$, $\dot{H}_t := \partial_\sigma H(R^\sharp_{\theta,\sigma}\rho_t)$,

$$\mathcal{A}_{\theta,\sigma}[\mathbf{x}] \;=\; \frac{1}{\theta^2}\sum_t \Gamma_t^2 \;-\; \frac{1}{\sigma^2}\sum_t \dot{H}_t \;+\; o(1)$$

as $(\theta,\sigma) \downarrow 0$. *If* $k = k^\natural$ *and* $\sigma^2 = 2Dh$ *with* $h \to 0$, *this reduces to the classical continuous Onsager–Machlup functional.*

Remark 3.2 (Data-driven OM). Any consistent drift estimate (e.g. LWLS–1) may replace m_θ without changing the quadratic structure of $\mathcal{A}_{\theta,\sigma}[\mathbf{x}]$.

3.5 Deterministic limit and least–action equivalence

Remark 3.3 (Classical Lagrangian limit). In the joint limit $\theta, h \to 0$ with $\sigma^2 = 2Dh$, the discrete action density converges to $\frac{1}{4D}\int_0^T \|\dot{x}(t) - b(x(t))\|^2 dt$, the standard Onsager–Machlup Lagrangian for diffusions.

Theorem 3.2 (KL–least–action equivalence)**.** *For every Rose kernel* $k_{\theta,\sigma}$ *let*

$$J(\theta,\sigma) := \mathrm{KL}\big(\mu_1 \,\|\, k_{\theta,\sigma}\#\mu_0\big), \qquad \mathcal{A}^{\mathrm{pop}}_{\theta,\sigma} := \mathbb{E}_{\mu_0}\big[\mathcal{A}_{\theta,\sigma}[X_{0:n}]\big]$$

denote, respectively, the population Kullback–Leibler functional and the population Onsager–Machlup action introduced in Corollary 3.2. Under Assumptions (K0)–(K3)

$$\boxed{\nabla_{(\theta,\sigma)} J \;=\; \nabla_{(\theta,\sigma)}\mathcal{A}^{\mathrm{pop}}_{\theta,\sigma}} \quad \textit{for all } (\theta,\sigma) \in (0,\infty)^2.$$

Remark 3.4. The pair

$$(\theta^\star, \sigma^\star) := \arg\min_{(\theta,\sigma)} J(\theta,\sigma) = \arg\min_{(\theta,\sigma)} \mathcal{A}^{\mathrm{pop}}_{\theta,\sigma}$$

maximises the likelihood of the observed path ensemble within the two-dial Rose family; equivalently, it minimises the expected Onsager–Machlup action. Hence trajectories generated under the little-rose kernel are the most probable (in the maximum-likelihood sense) among all trajectories that can be produced by Rose kernels.

3.6 Introducing the dial plane

(K4) C^1 regularity in dials. $(\theta,\sigma) \mapsto \log k_{\theta,\sigma}$ is C^1 and $\partial_\theta \log k_{\theta,\sigma}, \partial_\sigma \log k_{\theta,\sigma} \in L^2(d\mu_0 d\mu_1)$.

3.7 Edge-wise dial oracle

Oracle axioms

(A1) *Reversibility consistency.* If $k(y \mid x) = k(x \mid y)$ then $\mathcal{O}(x_i, x_j) = \mathcal{O}(x_j, x_i)$.

(A2) *Permutation equivariance.* $\mathcal{O}(\pi(i), \pi(j)) = \mathcal{O}(x_i, x_j)$ for any permutation π.

(A3) *Strict-convex minimal risk.* $\mathcal{O}(x_i, x_j) = \arg\min_{(\theta,\sigma)} \mathcal{J}_{ij}(\theta, \sigma)$ with $\mathcal{J}$ strictly convex.

(A4) Positive empirical mass. Empirical anchor weights satisfy $\inf_{i \in V} \widehat{\mu}_0(i) > 0$.

Definition 3.4 (Edge KL functional)**.** Given observed transitions from anchor x_i, let $\widehat{k}_{ij}$ denote the empirical conditional probability that a step from x_i lands in x_j. Define the edge-wise KL at dials (θ, σ):

$$\mathcal{J}_{ij}(\theta, \sigma) := \sum_j \widehat{k}_{ij} \log \frac{\widehat{k}_{ij}}{k_{\theta,\sigma}(x_j \mid x_i)}.$$

(For continuous X, replace the sum by an integral against the empirical conditional measure $\widehat{k}_i$.)

Proposition 3.2 (Edge-wise KL gradient)**.** *For every ordered pair (x_i, x_j) and all (θ, σ),*

$$\nabla_{(\theta,\sigma)} \mathcal{J}_{ij}(\theta, \sigma) = -\frac{1}{2} \nabla_{(\theta,\sigma)} \mathcal{I}_{ij}(\theta, \sigma).$$

Consequently the oracle dials $(\theta^\star_{ij}, \sigma^\star_{ij})$ maximise the Fisher load and minimise the edge-wise KL.

Definition 3.5 (Little-rose operator)**.** Assign each edge its oracle dials $(\theta^\star_{ij}, \sigma^\star_{ij})$ and denote by $\mathcal{R}^{\flat/\sharp}_\star$ the resulting pull-back and push-forward operators. We call this pair the *little-rose* operator.

3.8 Global KL gradient corollary

Corollary 3.3 (Global KL gradient)**.** *Summing Proposition 3.2 over all edges with weights $\mu_0(i)$ yields*

$$\nabla_{(\theta,\sigma)} \mathrm{KL}\big(\mu_1 \parallel k_{\theta,\sigma}\#\mu_0\big) = -\tfrac{1}{2} \nabla_{(\theta,\sigma)} \mathcal{I}_{\theta,\sigma},$$

hence the little-rose operator is the unique minimiser of KL and maximiser of the Fisher load.

4 Finite-sample dial selection

4.1 Goldenshluger–Lepski selector

Lemma 4.1 (Bias–variance decomposition)**.** *For either PF or Koopman risk,*

$$\mathbb{E}\, \widehat{\mathcal{R}}^{\square}(\theta, \sigma) \;=\; \mathrm{Bias}^2_{\square}(\theta, \sigma) \;+\; \mathrm{Var}_{\square}(\theta, \sigma),$$

where the bias term is the squared L^2 distance between the population Rose operator at (θ, σ) and the "oracle" operator that best fits the data class, and the variance term scales as

$$\mathrm{Var}_{\square}(\theta, \sigma) \;\lesssim\; \frac{\mathrm{trace}\big(\Sigma^{\square}_{\theta,\sigma}\big)}{n_{\mathrm{eff}}(\theta)},$$

with an effective sample size $n_{\mathrm{eff}}(\theta)$ determined by the local weight w_θ (see Appendix **??***).*

Theorem 4.1 (GL selector: non-asymptotic risk). *Let $\widehat{\theta}, \widehat{\sigma}$ be chosen by the GL rule with penalty parameter α. Then with probability at least $1-\delta$,*

$$\mathcal{R}\big(\widehat{\theta}, \widehat{\sigma}\big) \;\le\; C_\alpha \inf_{(\theta,\sigma)\in\mathcal{G}} \mathcal{R}(\theta,\sigma) \;+\; C\frac{\log(1/\delta)}{n},$$

*where $\mathcal{R}$ denotes either PF or Koopman L^2-risk and C_α, C are explicit constants (see Appendix **??**).*

Definition 4.1 (Sample risks). Let $\widehat{R}^{\sharp}_{\theta,\sigma}$ and $\widehat{R}^{\flat}_{\theta,\sigma}$ denote the empirical PF and Koopman estimators built from the sample. Given a test class $\mathcal{F} \subset L^2(X,\mu)$ and $\mathcal{P} \subset L^1(X,\mu)$ define

$$\widehat{\mathcal{R}}^{\mathrm{PF}}(\theta,\sigma) := \sup_{\rho\in\mathcal{P}} \big\|\widehat{R}^{\sharp}_{\theta,\sigma}\rho - R^{\sharp}_{\theta,\sigma}\rho\big\|^2_{L^2}, \qquad \widehat{\mathcal{R}}^{\mathrm{K}}(\theta,\sigma) := \sup_{f\in\mathcal{F}} \big\|\widehat{R}^{\flat}_{\theta,\sigma}f - R^{\flat}_{\theta,\sigma}f\big\|^2_{L^2}.$$

5 Directed curvature and reversibility diagnostics

5.1 Curvature proxy on the dial graph

5.2 Irreversibility measure via antisymmetric current

Definition 5.1 (Antisymmetric weight & current). For a Rose kernel $k_{\theta,\sigma}$ define the log–ratio antisymmetry

$$A_{ij} := \frac{1}{2}\Big(\log k_{\theta,\sigma}(x_j \mid x_i) - \log k_{\theta,\sigma}(x_i \mid x_j)\Big)$$

Given a reference mass $\pi_i > 0$ on vertices (e.g. stationary or empirical mass), the probability current is

$$J_{ij} := \pi_i k_{\theta,\sigma}(x_j \mid x_i) - \pi_j k_{\theta,\sigma}(x_i \mid x_j),$$

satisfying $J_{ij} = -J_{ji}$. Detailed balance holds iff all $A_{ij} = 0$ (equivalently $J_{ij} = 0$).

Lemma 5.1 (Graph Helmholtz split). *Let J be an antisymmetric edge flow on a finite connected graph (V,E) with vertex weights π. Then there exists a potential $\phi : V \to \mathbb{R}$ and a cycle flow $h \in \ker B^\top$ (where B is the weighted incidence matrix) such that*

$$J_{ij} = \pi_i k^{\natural}_{ij}(\phi_j - \phi_i) + h_{ij},$$

with $\sum_j \pi_i k^{\natural}_{ij}(\phi_j - \phi_i) = 0$ at stationarity. The decomposition is unique up to an additive constant in ϕ. Moreover $h = 0$ iff detailed balance holds.

Remark 5.1 (Loschmidt symmetry and the Fisher clock). Setting every antisymmetric weight to zero, $A_{ij} = 0$, collapses the kernel to its neutral form $k^{\natural}$ and makes the Rose operator exactly time–reversal symmetric, matching Loschmidt's objection to irreversible entropy growth.

Turning on any $A \neq 0$ injects a directed cycle curvature $F = \sum_{\mathrm{cycle}} A$, and the Fisher–curvature–entropy bridge (Thm. 3.1) yields

$$F \;=\; \frac{1}{2}\mathcal{I}_{\theta,\sigma} \;=\; \text{entropy production per step.}$$

A non-zero curvature therefore acts as an intrinsic *Fisher clock*—time emerges only once detailed balance is broken. In the Loschmidt-symmetric limit $A \to 0$ the clock halts and the macroscopic arrow of time disappears.

6 Discussion and Outlook

6.1 Key implications

(I1) Unified diagnostic of fluctuation strength
The Fisher load $\mathcal{I}_{\theta,\sigma}$ is *simultaneously* the variance coefficient, the synthetic–curvature prefactor, and the entropy–production rate. A single dial pair (θ, σ) governs them all.

(I2) Automatic bias–variance balance
Steepest descent of $\mathrm{KL}(\mu_1 \parallel k_{\theta,\sigma}\#\mu_0)$ is steepest *ascent* of $\mathcal{I}_{\theta,\sigma}$, so the statistically most informative kernel is found with no cross-validation.

(I3) One-step Schrödinger bridge
The little-rose kernel realises the discrete entropic transport $\mu_0 \to \mu_1$, linking operator learning to optimal-transport theory.

(I4) Landau lens on data
The Legendre transform of the CGF equips any observable with a Landau-type free-energy landscape whose stiffness is data-computable.

(I5) Lag–dimension selection. Remark **??** suggests using the little-rose Fisher load as a principled criterion for choosing delay-embedding parameters.

6.2 Future directions

(F1) Directed discrete curvature
Dial matrices $(\Theta_{ij}, \Sigma_{ij})$ define a doubly-weighted digraph, opening analytic work on Ollivier/Bakry–Émery curvature for non-reversible kernels.

(F2) Multiparameter persistent homology
The two weights furnish a natural bifiltration, inviting multi-parameter TDA of dynamical data sets.

6.3 Limitations

6.4 Conclusion

A Proof details for Sec. 3

B GL selector derivations

C Additional experiments

D Dial-graph topology (exploratory)